

Digital Design & Verilog

The skill behind VLSI jobs · Equigrow short notes

Combinational logic

```
module mux2 (input a, b, sel, output y);  
    assign y = sel ? b : a;  
endmodule
```

Sequential logic

```
module counter (input clk, rst, output reg [3:0] q);  
    always @(posedge clk) begin  
        if (rst) q <= 0;  
        else    q <= q + 1;  
    end  
endmodule
```

- Use <= (non-blocking) in clocked always blocks, = (blocking) in combinational ones.
- wire = connection; reg = holds a value inside always blocks.

Must-knows

- Number systems, Boolean algebra, K-maps, flip-flops (D, JK, T), setup and hold time.
- FSMs: Moore (output from state) vs Mealy (output from state + input).
- Always write a testbench to simulate before synthesis.